THE DRAWING SET: every sheet, in order

no	drawing	title	status	qty
1	DS-000	Index of the drawing set	PROPOSED	
2	GA-001	General arrangement	PROPOSED	1
3	HUB-200	Hub, locked, in its coupler	LOCKED	1
4	HUB-201	Rings A and B, Cu foil, beaded	LOCKED	
5	HUB-202	Vessel, borosilicate, Ø50 × 1.5	LOCKED	1
6	HUB-203	Retainer, PEEK	PROPOSED	1
7	HUB-204	Shaft coupler, G10 tube 60 / 66	PROPOSED	1
8	HUB-205	AH former, G-10	LOCKED	2
9	HUB-206	AH coil, 160 turns	LOCKED	2
10	SHF-300	Shaft half A with its flange, sleeves and sheds	PROPOSED	1
11	SHF-301	Shaft half A, Ø25, its LV bore	PROPOSED	1
12	SHF-302	Shaft half B, Ø25, its LV bore	PROPOSED	1
13	SHF-303	Shaft-half flange, Ø60 × 8	PROPOSED	2
14	SHF-304	Rotor sleeve, the vane bay, G10	PROPOSED	2
15	SHF-305	Rotor sleeve, the reluctance bay, G10 (A; B mirror)	PROPOSED	2
16	SHF-306	PTFE cover over the sleeve	PROPOSED	2
17	SHF-307	PTFE shed, Ø65 × 2	PROPOSED	4
18	CAN-400	Can over the hub, with its end discs and bushings	PROPOSED	1
19	CAN-401	Can, Al, Ø96 × 2 × 170	PROPOSED	1
20	CAN-402	Can end disc, three M4 holes	PROPOSED	2
21	CAN-403	PTFE bushing, the ring's lead	PROPOSED	2
22	DCK-500	Deck A, its parts potted on the G10 plate	PROPOSED	2
23	DCK-501	Deck A wiring list	PROPOSED	2
24	DCK-502	Deck plate, G10, r 50-150 × 6	PROPOSED	2
25	DCK-503	Junction terminal in its potted block	PROPOSED	20
26	DCK-504	Module: HV rectifier stick, potted	PROPOSED	12
27	DCK-505	Module: surge resistor, potted	PROPOSED	4
28	DCK-506	Module: capacitor C3-C6, potted	PROPOSED	4
29	DCK-507	Module: capacitor C7 / C8, potted	PROPOSED	2
30	DCK-508	Module: capacitor Csa / Csb, potted	PROPOSED	2
31	DCK-509	Module: clamp string Z1 / Z4, potted	PROPOSED	2
32	CAP-600	C1 / C2 varicap stack, side A	PROPOSED	2
33	CAP-601	Rotor vane, C1 / C2 (r03-r11)	PROPOSED	10
34	CAP-602	Rotor vane, the first (r01), hands A / B	PROPOSED	2
35	CAP-603	Stator vane with its six lugs, C1 / C2	PROPOSED	12
36	CAP-604	Rotor-vane spacer ring, G10	PROPOSED	10
37	CAP-605	Dowel, G10, Ø5 × 93	PROPOSED	6
38	CAP-606	Stator spacer tube, G10, Ø16 / 8.5 (-01 × 15, -02 × 12)	PROPOSED	72
39	CAP-607	End ring, G10, in the cage's bore	PROPOSED	2
40	CAP-610	Ca / Cb column, side A: the two combs on their posts	PROPOSED	2
41	CAP-611	Ca plate, node-1 comb (Ca_03, Ca_01)	PROPOSED	4
42	CAP-612	Ca plate, node-1 comb's end (Ca_05)	PROPOSED	2
43	CAP-613	Ca plate, node-2 comb (Ca_04)	PROPOSED	2
44	CAP-614	Ca plate, node-2 comb's end (Ca_02)	PROPOSED	2
45	CAP-615	Ca plate, node-2 comb's base (Ca_06), hands A / B	PROPOSED	2
46	CAP-616	Post spacer, Al, Ø8, M4 (-01 × 15, -02 × 6)	PROPOSED	30
47	CAP-617	Standoff, glazed steatite, Ø16 × 30, M4	PROPOSED	6
48	REL-700	Utron, wound: half-cores, neck strip, coil	LOCKED	6
49	UTR-101	Half-core, laminated (hand A; hand B mirror)	LOCKED	
50	REL-702	Neck strip, 80 % NiFe	LOCKED	6
51	REL-703	Air-break spacer, G10	LOCKED	6
52	REL-704	Slot cover, G10, bonded	LOCKED	6
53	REL-705	Cheek, G10 (-01 +v drawn; -02 -v the mirror)	LOCKED	24
54	REL-706	Yoke coil, 200 turns	LOCKED	6
55	REL-710	Reluctance section A: utrons, carrier discs, bridges, bridge ring	LOCKED	2
56	REL-711	Utron carrier disc, G10	LOCKED	4
57	REL-712	Bridge, M235-35A stack	LOCKED	12
58	REL-713	Bridge ring, G10 (A; B the mirror)	LOCKED	2
59	REL-720	La / Lb choke, EI-84 × 35, 150 turns	PROPOSED	2
60	REL-721	La / Lb winding, 150 turns of Ø1.40	PROPOSED	2
61	REL-730	Start kick K4, module	PROPOSED	1
62	BRG-800	Bearing hub, Ca|reluctance, side A	PROPOSED	2
63	BRG-801	Bearing hub, end (frame), side A	PROPOSED	2
64	BRG-802	Bearing hub, the bridge ring's, side A	PROPOSED	2
65	BRG-810	Spider, Ca|reluctance, G10, Ø364	PROPOSED	2
66	BRG-811	Spider, end (frame), G10, Ø324	PROPOSED	2
67	BRG-812	Spider, the bridge ring's bearing, G10	PROPOSED	2
68	BRG-813	Ring adapter, G10, spider to bridge ring	PROPOSED	2
69	BRG-814	Cartridge, the bridge-ring bearing's (-01 A; -02 B the mirror)	PROPOSED	2
70	CGE-900	Counter-rotor: the cage, the stator stacks, the spiders, the bridges	PROPOSED	1
71	CGE-901	Cage tube, G10, Ø364 / 372	PROPOSED	1
72	DRV-100	The 1 : −1 bevel reverser, side A	PROPOSED	1
73	DRV-101	Side gear, the rotor's, z 30 m 2, keyed	PROPOSED	1
74	DRV-102	Side gear, the counter-rotor's, z 30 m 2, flanged	PROPOSED	1
75	DRV-103	Pinion, POM-C, z 20 m 2	PROPOSED	3
76	DRV-104	Stub shaft, Ø10, the pinion's	PROPOSED	3
77	DRV-105	Cartridge sleeve, the pinion's two 6000-2Z	PROPOSED	3
78	DRV-106	Gear case, the carrier and end-bearing housing	PROPOSED	1
79	DRV-107	Bell, a C under the case floor	PROPOSED	1
80	DRV-108	Counter-rotor hub, the gear-end bearing's	PROPOSED	1
81	DRV-109	End disc, G10, the counter-rotor's	PROPOSED	1
82	DRV-120	The belt drive: pulleys, belt, motor on its plate	PROPOSED	1
83	DRV-121	Motor plate, tensioning slots	PROPOSED	1
84	DRV-190	The drive's spacers, rings, keys and fasteners	PROPOSED	
85	P-001	Bearing, SKF 6205-2Z	BOUGHT	6
86	P-002	HV rectifier stick, 20 kV, 5 mA (XCH 2CL77)	BOUGHT	12
87	P-003	HV capacitor, 190 pF / 30 kV, N4700 (Vishay 715C30KTT19)	BOUGHT	4
88	P-004	HV capacitor, 100 pF / 40 kV, N4700 (Vishay 715C40KTT10)	BOUGHT	4
89	P-005	Surge resistor, 22 kΩ, ≥ 15 kV, ≥ 1 W	BOUGHT	4
90	P-006	TVS diode 1.5KE200A (the clamp strings)	BOUGHT	132
91	P-007	MnZn ferrite rod, Fair-Rite 4077484611 (77)	BOUGHT	2
92	P-008	Threaded rod M8 and nuts, glass-epoxy	BOUGHT	12 + 24
93	P-009	HV wire, FEP, 20 kV, 18 AWG (UL3239)	BOUGHT	—
94	P-010	LV cable, the magnetic pump's leads	BOUGHT	2
95	P-011	Fasteners, inserts and stock	BOUGHT	
96	P-012	Bearing, SKF 6000-2Z	BOUGHT	6
97	P-013	Timing belt, Gates 750-5M-15 (HTD 5M)	BOUGHT	1
98	P-014	Pulleys HTD 5M: 80T taper-bush, 20T flanged	BOUGHT	1 + 1
99	P-015	Motor, BLDC 200 W 48 V (APBA60M048030-E), with driver	BOUGHT	1
100	P-016	Schottky rectifiers D1*-D4*, DO-201AD	BOUGHT	2 + 2
101	P-017	Node snubbers: 22 nF PP film + 240 Ω (8 of each)	BOUGHT	8 + 8
102	P-018	AH bypass capacitor, 22 mF 10 V, snap-in	BOUGHT	2
103	P-019	Choke parts: EI-84 laminations, bobbin, wire, spacer	BOUGHT	2 sets

NOTES

1 The set draws the machine as sim/tube_geometry.py LayoutMachine builds it (1014 mm; 552 solids from 168 prototypes): the record's locked design, the multiplier's layout (PROPOSED, sim/multiplier-layout-findings.md), the bridge-ring bearings (PROPOSED, sim/bridge-ring-bearings-findings.md) and the capacitors' mounts (PROPOSED, sim/doubler-caps-findings.md).

2 Status per sheet: LOCKED, the record's locked design (docs/ledger/DCCREG-design-ledger.md); PROPOSED, a proposal for the designer to accept; BOUGHT, a source-control sheet for a bought part.

3 Every sheet is Rev A, a draft for quotation: the drawings are generated from the solids and are to be checked and approved before release. Side B's parts are side A's mirror or the same part, as each sheet states.

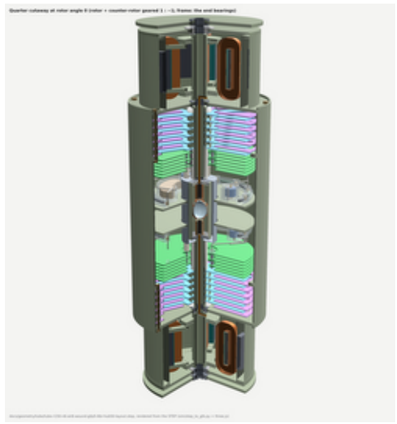
4 The proposed parts the layout's model does not carry are drawn from their first cuts (sim/unmodelled_parts.py): the magnetic pump's chokes, kick and electronics (REL-720 to REL-730, P-016 to P-019; their places on the rotor OPEN) and the drive (DRV-100 to DRV-190, P-012 to P-015; on the record's 940 mm stack-up, to be re-cut on the layout: ledger §6, item 71).

5 Not drawn: the screws, studs and inserts (sheets P-011 and DRV-190 list them), the gel, the potting moulds, the frame (the base plate, the columns, the top plate: placeholders in docs/drive-gear-belt.md §9).

REVISIONS

Rev	Date	Description
A	2026-10-11	First issue, draft for quotation

status	PROPOSED
qty per machine	—
used in	
source	
sheets	the index and every drawing below
model	docs/geometry/tube/…-layout.step
generator	docs/make_drawing_set.py



the machine as the set draws it (the layout, quarter cut)

DCCREG turbine · the drawing set

Index of the drawing set		Drawing no.
Material:		**DCCREG-DS-000**
Mass: —	Qty per machine: —	Rev A Sheet 1 of 103 A3
Scale: —	Tolerances: ISO 2768-mK	Date: 2026-10-11
Drawn: generated by docs/make_drawing_set.py	Checked: — Approved: —	**PROPOSED · DRAFT**